



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Let's read about software programming's history.

```
function updatePhotDescription(){  
    if (descriptions.length> (page*9)  
    .  
    }  
}  
  
function updatePhotDescription(){  
    if (descriptions.length> (page*9)  
    .  
    }  
    var i = 1;  
    while (i< 10){  
        var elementId= foto+ i;  
        var element.getById(elementBig)  
    } else {
```

Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.

Give detailed and important information regarding to the history of language programming in different periods of time.

Ask your partner:

- What do you know about the history of computers?
- How can you compare computers when you were a child and computers you are using now?

Can Do Statements:

- I can give important information related to the history of software programming and its evolution.
- I can talk about the fundamental paradigms in software programming.
- I can identify the fundamental paradigms in software programming.





Let's Explore!

Jane and Jacob are talking about fundamental paradigms in software programming.



Instructions: Read the text below.

Jacob: Hi, Jane. Do you have a minute?

Jane: Hi, Jacob. Sure, no problem. How can I help you?

Jacob: I have problems with the history of software programming and its evolution. I used to know about it, but now I can't remember. Did you learn something about it back in school?

Jane: Actually, I did. Let's see... The first computer program was the Analytical Engine, it was designed by Charles Babbage who also introduced very functional and important computing concepts. Then, Herman Hollerith provided an input and output using a very practical method: punched cards. At the beginning of the 20th century, the first computers had specialized codes for their applications. Numerical calculations were based on decimal numbers, which meant that logic was represented with both, number and words.

Jacob: Did you find information about the first programming languages?

Jane: Well, that's not precise. There is some information dated back in 1940 when computers were electrical and included limited speed and memory capacity. In the 1950's, the Germans invented a high-level programming language that communicated instructions to computers. This decade was considered a starting point where the first steps appeared for programming language.

Jacob: So, did this period establish the fundamental paradigms for programming languages?

Jane: Not exactly. The 60s and the 70s brought a major flowering in the field of programming languages. For example, most of the language paradigms created in that time such as Speakeasy, Simula, Smalltalk, and Prolog are still in use. What we need to know is that these paradigms became the stepping stones for the current programming languages.

Jacob: Jane, you really know about this. Thank you!

Jane: No problem! Anytime.





Let's Practice!

Software Programming and Its Evolution

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

Word Bank

Analytical Engine / numerical calculations / decimal numbers

1. The history of software programming starts with the _____ as the first computer program ever created.
2. At the beginning, _____ were based on decimal numbers.
3. Throughout the time, numerical calculations had an evolution, and now they do not just use _____ but mostly binary codes.



Let's Practice!

Identifying the Fundamental Paradigms in Software Programming

Instructions: Select the best options in the table. There is only one answer per sentence.

	Speakeasy	Smalltalk	Simula	Prolog
Did this language paradigm support the first object-oriented programming? Which was it?				
Did this language paradigm support the first object-oriented programming? Which was it?				
Did this language paradigm provide a ground-up design for an object-oriented programming?				
Did Colmerauer, Roussel, and Kowalski create this logic programming as the first one ever?				



Extension Activity

Fundamental Paradigms in Software Programming

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about fundamental paradigms in software programming.
- Create a post with the main characteristics of fundamental paradigms in software programming.
- Share with your partners the post you created.

Self-Assessment

Yes

No

- I look for more information about fundamental paradigms in software programming. ☐ ☐
- I create a post with the main characteristics of fundamental paradigms in software programming. ☐ ☐
- I share with your partners the post you created. ☐ ☐



Target Words

- **stepping stone** (expression): a situation that helps to approach an objective
e.g. What we need to know is that these paradigms became the stepping stones for the current programming languages.
- **paradigm** (noun): a model, a pattern
e.g. For example, most of the language paradigms created in that time such as Speakeasy, Simula, Smalltalk, and Prolog are still in use.
- **punched card** (noun): a card in which data can be coded by making holes
e.g. Then, Hernan Hollerith provided an input and output using a very practical method: punched cards.
- **flowering** (adjective): growing, increasing
e.g. The 60s and the 70s brought a major flowering in the field of programming languages.
- **decade** (noun): a period of ten years
e.g. This decade was considered a starting point where the first steps appeared for programming language.
- **current** (adjective): present time
e.g. What we need to know is that these paradigms became the stepping stones for the current programming languages.



Learning Points

Adjectives Ending In "al"

Adding a suffix to some verbs and nouns is one way to form adjectives.

There are some latin-rooted nouns and verbs that can become adjectives by adding the suffix -al. See the following charts:

Nouns	Adjectives
practice	<i>practical</i>
function	<i>functional</i>
element	<i>elemental</i>
structure	<i>structural</i>
fundament	<i>fundamental</i>

Verbs	Adjectives
add	<i>additional</i>
intent	<i>intentional</i>
benefit	<i>beneficial</i>
converse	<i>conversational</i>
criticize	<i>critical</i>

When using these adjectives in a sentence, place them in the usual adjective positions. For example:

- *Programming nowadays is more **functional** than in the past.*
- ***Structural** problems need to be solved carefully.*

Examples

- *The first computer program was the Analytical Engine, it was designed by Charles Babbage who also introduced very functional and important computing concepts.*
- *Then, Hernan Hollerith provided an input and output using a very practical method: punched cards.*
- *Numerical calculations were based on decimal numbers, which meant that logic was represented with both, number and words.*
- *There is some information dated back in 1940 when computers were electrical and included limited speed and memory capacity.*





Learning Points

Simple Past Tense in Questions

You use **the simple past tense** form to talk about actions that started and finished in the past.

When using the simple past tense, you can use time expressions to indicate when the action took place.

Examples:

*I **took** the test about fundamental paradigms **last June**.*

*Michelle **designed** the logo in **December 1999**.*

Remember that the verbs in simple past tense can be regular or irregular. Regular verbs are the ones that end in **-ed**, **-ied** or **-d**. On the other hand, irregular verbs are the ones that have different variations when used in the simple past.

Examples:

*My boss **bought** me a new computer.*

*In 1950, Germans **invented** a high-level programming language.*

When asking **questions in simple the simple past tense**, you use the auxiliary **did**. However, remember that the action verb keeps its base form. Check these examples:

***Did** you **buy** the balloons for the party?*

***Did** the Germans **invent** a high-level programming language?*

Examples

- *Did you learn something about it back in school?*
- *Did you find information about the first programming languages?*
- *Did this period establish the fundamental paradigms for programming languages?*

